N078AV (S/N: W20051877422) — NVMe SSD Hardware Failure Report

Immfly Ground Operations / IT Session: September 21, 2026

Field	Value
Machine	**N078AV** (S/N: W20051877422)
Affected component	Primary Solid State Drive (`/dev/nvme0n1`) — M.2 NVMe Storage Unit
Initial symptom	Continuous block layer medium read aborts flooding console at login stage: `blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ)`.
Hardware status	**HARDWARE FAILURE** — Physical NAND flash bad sector on device `nvme0n1` at sector `1445656` (LBA 1,445,656). Unrecoverable ECC failure.
Root cause	Permanent hardware cell degradation and depletion of reallocated spare sectors within the primary NVMe SSD. The internal flash memory controller encounters an uncorrectable bit-error rate (ECC failure) on sector 1445656 and repeatedly returns hardware aborts to the Linux kernel block layer.
Comparison	Shares identical root symptoms with N067AV (NVMe medium error on sectors 791664640+) and N084AV (early boot sector failure on sector 0x79aee0). While N078AV reached the login prompt, the persistent I/O failure renders filesystem operations unstable, causing eventual kernel lockups, read-only remounts, and service downtime.
Recommended action	**Physical NVMe SSD replacement required.** Complete operating system re-imaging and fleet content package reload.

NARRATIVE SUMMARY

During operational execution and system startup on aircraft server unit **N078AV (S/N: W20051877422)**, th host kernel console was persistently interrupted by repeated block layer I/O aborts targeting the primary NVM storage drive (`dev nvme0n1`).

Specifically, the Linux kernel reported identical read failure sequences: `blk_update_request: critica medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags 0x0 phys_seg 1 pri class 0`, recurring deterministically every ~5.25 seconds between timestamps `106.81s` and `217.06s`.

Under the NVM Express (NVMe) specification, a **Critical Medium Error** (Status Code `0x0282` - Unrecovere Read Error) indicates physical degradation of flash memory cells where internal low-density parity-chec (LDPC) or BCH Error Correction Code (ECC) algorithms can no longer restore data integrity. The SSI controller cannot service requests for logical block address (LBA) `1445656` nor transparently reallocate th sector from internal reserve pools.

Although the system managed to initialize the serial login prompt (`p100-ify login:`), any backgroun service, Docker container, or database transaction attempting to read inodes or data blocks spanning secto `1445656` immediately receives fatal `EIO` (Input/output error). The storage unit is physically degraded an cannot maintain runtime stability.

N078AV (S/N: W20051877422) — Diagnostic Timeline & Console Appendix

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DIAGNOSTIC FINDINGS & SEQUENCE OF EVENTS

1. **Kernel Block Layer Interrogation:**
   Following system bootstrap, userland background processes or page cache flushes triggered read requests across primary storage block device `/dev/nvme0n1`.
2. **Persistent Read Faults on Sector 1445656:**
   Beginning at timestamp `106.816s`, each asynchronous page read on sector `1445656` failed unconditionally. Over 22 continuous iterations were logged through `217.068s`.
3. **Controller-Level Media Rejection:**
   The repeated recurrence of the exact same sector fault demonstrates that the NVMe drive controller is unable to read or remap the bad block, confirming that flash spare block reserves are exhausted or hardware controller firmware has failed to isolate the damaged block.
4. **Filesystem Risk & Operational Degradation:**
   Sectors in this lower range (LBA 1,445,656 ≈ 705 MB offset) correspond to core operating system binaries, shared libraries, or filesystem superblock/inode tables. Continued operation will trigger read-only remounts (`remount-ro`), container crash loops, or kernel panics upon execution of affected sectors.

CONCLUSION AND RECOMMENDATION

The primary NVMe SSD installed in server **N078AV** has suffered unrecoverable physical hardware degradation. Software-based repair utilities (such as `fsck` or `badblocks`) cannot resolve underlying NAND silicon degradation or hardware controller read aborts.

Recommended action: Issue an urgent Line Maintenance Work Order to **physically replace the NVMe Solid State Drive (M.2 SSD)** in aircraft server unit **N078AV (S/N: W20051877422)**, followed by operating system provisioning and content package synchronization.

APPENDIX: CONSOLE BOOT FAILURE OUTPUT

```
[BOOT:N078AV] Kernel NVMe block layer critical medium errors at console login:
p100-ify login: [  106.816056] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[  112.068228] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  117.325789] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  122.574875] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  127.823516] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  133.074806] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  138.317653] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  143.576878] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  148.824184] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  154.068000] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  159.323363] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  164.572687] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
```

```
0x0 phys_seg 1 prio class 0
[  169.819138] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  175.070972] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  180.323858] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  185.576044] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  190.823552] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  196.001838] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  201.321956] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  206.574029] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  211.823640] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
[  217.068947] blk_update_request: critical medium error, dev nvme0n1, sector 1445656 op 0x0:(READ) flags
0x0 phys_seg 1 prio class 0
```